gpeyt1-15 · Piezometer QA

GEO PE · Long Calculation Problems I — Topic 3: Construction Observation, QA/QC & Safety

Problem

A vibrating-wire piezometer is installed at the middle of a soft clay layer (tip at El. 60.0) beneath a staged embankment. Immediately after stage-1 fill, the excess pore pressure at the tip was 1,150 psf. Three months later the piezometer indicates a total head equivalent to a water level at El. 96.5, while the static groundwater table is at El. 88.0. Settlement plates show 7.2 in. of settlement; inclinometers show <0.3 in. lateral movement.

Most nearly, the local degree of consolidation at the piezometer tip is:

A. 54%
B. 46%
C. 76%
D. 19%

Given equations / data

> ue = γw × (piezometric elevation − static water elevation); Uz = 1 − ue/ue0; γw = 62.4 pcf

Worked solution

1. Excess head = 96.5 − 88.0 = 8.5 ft
2. ue = 8.5 × 62.4 = 530 psf
3. Uz = 1 − 530/1,150 = 0.539 ≈ 54%

Final answer

54%

Why the other values are wrong

- **46%** — reported ue/ue0 (fraction still remaining)
- **76%** — measured excess head from the ground surface (El. 92.0) instead of the static water table
- **19%** — used the clay unit weight (110 pcf) instead of γw

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.11 Geotechnical Instrumentation, p. 156; §3.2.3 consolidation

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=SXA_TC3XEwk (no creator values copied).

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